10.

Between the Planets

From a distance, our solar system looks empty. If you enclosed it within a sphere—one large enough to contain the orbit of Neptune, the outermost planet[†]—then the volume occupied by the Sun, all planets, and their moons would take up a little more than one-trillionth the enclosed space. But it's not empty, the space between the planets contains all manner of chunky rocks, pebbles, ice balls, dust, streams of charged particles, and far-flung probes. The space is also permeated by monstrous gravitational and magnetic fields.

Interplanetary space is so not-empty that Earth, during its 30 kilometer-per-second orbital journey, plows through hundreds of tons of meteors per day—most of them no larger than a grain of sand. Nearly all of them burn in Earth's upper atmosphere, slamming into the air with so much energy that the debris vaporizes on contact. Our frail species evolved under this protective blanket. Larger, golf-ball-size meteors heat fast but unevenly, and often shatter into many smaller pieces before they vaporize. Still larger meteors singe their surface but otherwise make it all the way to the ground intact. You'd think that by now, after 4.6 billion trips around the Sun, Earth would have "vacuumed" up all possible debris in its orbital path. But things were once much worse. For a half-billion years after the formation of the Sun and its planets, so much junk rained down on Earth that heat from the persistent energy of impacts rendered Earth's atmosphere hot and our crust molten.

One substantial hunk of junk led to the formation of the Moon. The unexpected scarcity of iron and other higher-mass elements in the Moon, derived from lunar samples returned by Apollo astronauts, indicates that the Moon most likely burst forth from Earth's iron-poor crust and mantle